

for i in range(epochs):

for j in range(x.shape[0]):

- Select 1 row (random)
- Predict (using Forward prop)
- Calculate loss (using loss function → mse)
- Update weights and bias using GD

$$w_n = w_0 - \eta \frac{\partial L}{\partial w}$$

→ Calculate avg loss for the epoch
Loss

$$\frac{\partial L}{\partial w_{21}} = -2(y - \hat{y})a_{12}$$

$$\frac{\partial L}{\partial b_{21}} = -2(y - \hat{y})$$

$$\frac{\partial L}{\partial w_{11}} = -2(y - \hat{y})w_{11}x_{i1}$$

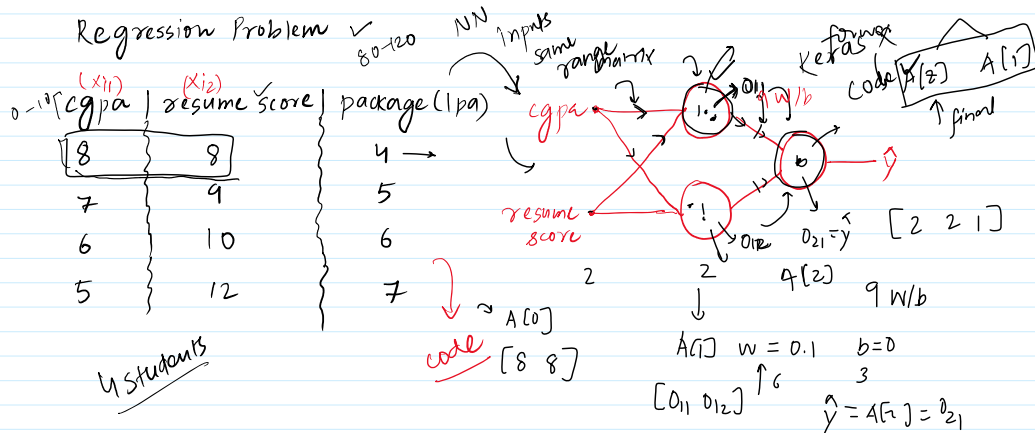
$$\frac{\partial L}{\partial w_{21}} = -2(y - \hat{y})w_{11}x_{i2}$$

$$\frac{\partial L}{\partial b_{11}} = -2(y - \hat{y})w_{11}$$

$$\frac{\partial L}{\partial w_{12}} = -2(y - \hat{y})w_{21}x_{i1}$$

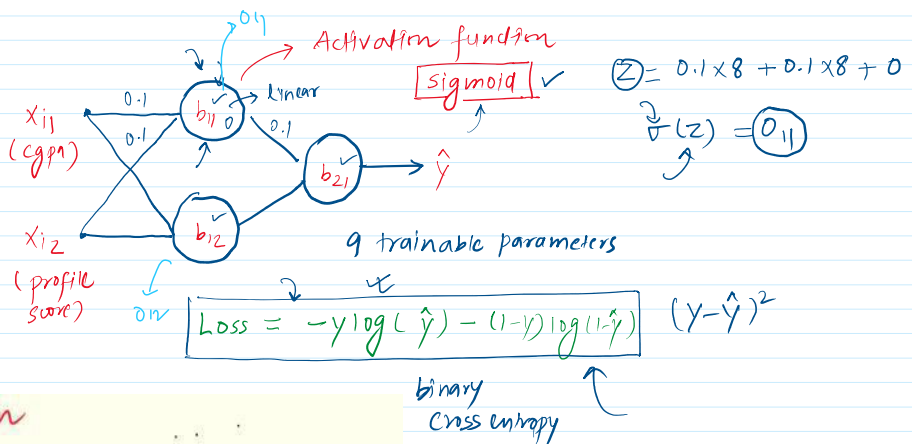
$$\frac{\partial L}{\partial w_{22}} = -2(y - \hat{y})w_{21}x_{i2}$$

$$\frac{\partial L}{\partial b_{12}} = -2(y - \hat{y})w_{21}$$



Classification Example

cgpa	profile score	placement
8	8	1
7	9	1
6	10	0
5	5	0



Backpropagation Algorithm

epochs = 5

for i in range(epochs):

for j in range(x.shape[0]):

- Select 1 row (random)
- Predict (using Forward prop)
- Calculate loss (using loss function)

bce

code convert

- > Here is the implemetation of an example of regression problem.
- > We implement this examle in code with different approach
- > first is raw code and second is using buildin library
- > and we get same result in each approach
- > the example code of campusx is ulploaded on the github

Back propagation Algorithm

epochs = 5 ✓

for i in range(epochs): (rows)

for j in range(x.shape[0]):

- Select 1 row (random) → 1by1
- Predict (using Forward prop) → predict
- Calculate loss (using loss function → bce)
- Update weights and bias using GD

code convert }

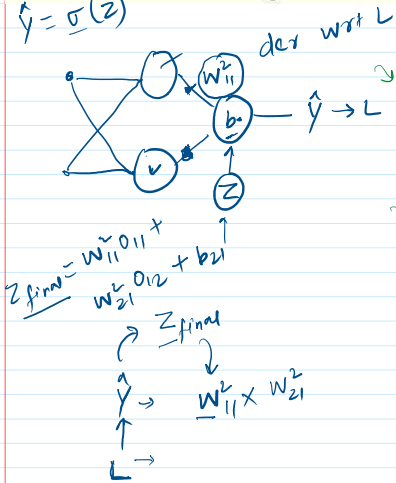
$$W_n = W_0 - \eta \left(\frac{\partial L}{\partial W} \right) \quad \text{q w, b update}$$

→ Calculate avg loss for the epoch
Loss

$$Z = W_{11}^2 \delta_{11} + W_{21}^2 \delta_{21} + b_{21}$$

$$\hat{y} = \sigma(z)$$

$$L = -y \log(\hat{y}) - (1-y) \log(1-\hat{y})$$



$$\frac{\partial L}{\partial W_{11}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial W_{11}^2} = \boxed{-(y-\hat{y}) \delta_{11}} \quad (3)$$

$$\frac{\partial L}{\partial W_{21}^2} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial W_{21}^2} = \boxed{-(y-\hat{y}) \delta_{12}}$$

$$\frac{\partial L}{\partial b_{21}} = \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} \times \frac{\partial z}{\partial b_{21}} = \boxed{-(y-\hat{y})}$$

$$\left[\frac{\partial L}{\partial \hat{y}} \right] = \frac{\partial}{\partial \hat{y}} [-y \log(\hat{y}) - (1-y) \log(1-\hat{y})]$$

$$= -\frac{y}{\hat{y}} + \frac{(1-y)}{(1-\hat{y})} = \frac{-y(1-\hat{y}) + \hat{y}(1-y)}{\hat{y}(1-\hat{y})} = \frac{-y + y\hat{y} + \hat{y} - y\hat{y}}{\hat{y}(1-\hat{y})}$$

$$= \boxed{\frac{-(y-\hat{y})}{\hat{y}(1-\hat{y})}} \quad \frac{\partial \sigma(z)}{\partial z} = \sigma(z) [1 - \sigma(z)]$$

$$\left[\frac{\partial \hat{y}}{\partial z} \right] = \frac{\partial (\sigma(z))}{\partial z} = \sigma(z) [1 - \sigma(z)] = \hat{y} (1-\hat{y})$$

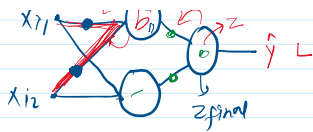
$$\frac{\partial L}{\partial \hat{y}} = \frac{-(y-\hat{y})}{\hat{y}(1-\hat{y})} \quad \frac{\partial \hat{y}}{\partial z} = \hat{y}(1-\hat{y}) \Rightarrow \frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z} = \frac{-(y-\hat{y})}{\hat{y}(1-\hat{y})} \times \hat{y}(1-\hat{y}) = -(y-\hat{y})$$

$$\frac{\partial z}{\partial W_{11}^2} = W_{11}^1 x_{11} + W_{21}^1 x_{12} + b_{11}$$

$$\downarrow -(y-\hat{y})$$



$$z_f = W_{11}^2 \delta_{11} + W_{21}^2 \delta_{21} + b_{21} \delta_{11} (1-\delta_{11})$$



$$\frac{\partial L}{\partial w'_{11}}, \frac{\partial L}{\partial w'_{21}}, \frac{\partial L}{\partial b_{11}}$$

$$z_f = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{21} o_{11} (1 - o_{11})$$

$$o_{11} = \sigma(z_{prev})$$

$$= \sigma(z_{prev}) [1 - \sigma(z_{prev})]$$

$$L \rightarrow \hat{y} \rightarrow z_f \rightarrow o_{11}$$

$$\downarrow$$

$$z_{prev}$$

$$\downarrow$$

$$w'_{11}$$

$$\frac{\partial L}{\partial w'_{11}} = \left[\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_f} \right] \times \left[\frac{\partial z_f}{\partial o_{11}} \times \frac{\partial o_{11}}{\partial z_{prev}} \times \frac{\partial z_{prev}}{\partial w'_{11}} \right]$$

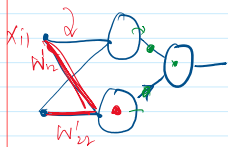
$$\downarrow \quad \downarrow \quad \downarrow$$

$$- (y - \hat{y}) \quad w_{11}^2 \quad o_{11} (1 - o_{11}) \quad x_{i1}$$

$$\frac{\partial L}{\partial w'_{11}} = - (y - \hat{y}) w_{11}^2 o_{11} (1 - o_{11}) x_{i1} \quad (3)$$

$$\frac{\partial L}{\partial w'_{21}} = - (y - \hat{y}) w_{11}^2 o_{11} (1 - o_{11}) x_{i2}$$

$$\frac{\partial L}{\partial b_{11}} = - (y - \hat{y}) w_{11}^2 o_{11} (1 - o_{11})$$



$$\frac{\partial L}{\partial w'_{12}}, \frac{\partial L}{\partial w'_{22}}, \frac{\partial L}{\partial b_{12}}$$

$$z_f = w_{11}^2 o_{11} + w_{21}^2 o_{12} + b_{21}$$

$$\downarrow$$

$$o_{12}$$

$$L \rightarrow \hat{y} \rightarrow z_f \rightarrow o_{12}$$

$$\downarrow$$

$$w_{12} \leftarrow z_p$$

$$\frac{\partial L}{\partial w'_{12}} = \left[\frac{\partial L}{\partial \hat{y}} \times \frac{\partial \hat{y}}{\partial z_f} \right] \times \left[\frac{\partial z_f}{\partial o_{12}} \times \frac{\partial o_{12}}{\partial z_p} \times \frac{\partial z_p}{\partial w'_{12}} \right]$$

$$\downarrow \quad \downarrow \quad \downarrow$$

$$- (y - \hat{y}) \quad w_{21}^2 \quad o_{12} (1 - o_{12}) \quad x_{i1}$$

$$\frac{\partial L}{\partial w'_{12}} = - (y - \hat{y}) w_{21}^2 o_{12} (1 - o_{12}) x_{i1} \quad (3)$$

$$\frac{\partial L}{\partial w'_{22}} = - (y - \hat{y}) w_{21}^2 o_{12} (1 - o_{12}) x_{i2}$$

$$\frac{\partial L}{\partial b_{12}} = - (y - \hat{y}) w_{21}^2 o_{12} (1 - o_{12})$$

Backpropagation → The Why?

The intuition behind the algorithm ✓

- > Here we have also implemented the same example for the classification problem.
- > Almost everything is same as before except the loss function and the activation function.
- > so the derivatives are also different here
- > We have calculated the 9 derivatives for this problem
- > and implement the full approach on code by raw and using libraries.
- > code is uploaded on github
- > <https://github.com/tanvirahmed1732/My-Learnings/tree/main/Deep%20Learning>